

Build Your Own Algorithm Autoresearch Framework



Welcome! This track is organised by C3 and AIDDA. C3 provides GPU and CPU compute for research groups and the agents that work for them, while AIDDA coordinates research, knowledge sharing, and networking around AI-driven algorithm discovery. We're excited to see what you build.

At a glance

- Build: An algorithm autoresearch framework
- Submit: Email admin@algorithmdiscovery.org with repo + max 4-minute video presentation
- Judging: Novelty, performance, interpretability, research efficiency
- Resources: Compute and LLM/API tokens are provided for the hackathon.
- Support: Join the event Discord <https://discord.gg/ZN36RsFPs>, find us at the event, or email admin@algorithmdiscovery.org.
- Prize: Track prizes + \$1000 of compute credits

The Challenge

Build an autoresearch framework: a system capable of automatically discovering, testing, and improving algorithms.

Your system should take a well-defined problem or class of problems and autonomously explore possible improvements, for example by

generating ideas or code, running experiments, analysing results, retaining useful discoveries, and proposing new approaches.

For inspiration, see systems such as Andrej Karpathy's Autoresearch and DeepMind's AlphaEvolve, as well as open/community autoresearch efforts such as the ECDSA.fail challenge.

What Can Your System Work On?

You are free to choose the area(s) in which your autoresearch system operates.

Possible examples include:

- [Mathematical solution discovery, e.g. Georgiev et al. \(2025\).](#)
[Mathematical exploration and discovery at scale](#)
- Operations Research and optimisation
- ML training or inference
- Systems and performance optimisation
- Scientific computing

Your framework does not need to work across every domain. It may be specialised to a particular field, for example, ML inference optimisation. But it should represent a reusable research system rather than being hard-coded to solve one specific benchmark.

What You Submit

Email your submission to admin@algorithmdiscovery.org with:

- Team name
- Repo URL
- Short description

- Link to video presentation (max 4 minutes, uploaded to Google Drive)

The repository should include:

- The source code for your system
- Clear instructions for installing and running it
- An explanation of the high-level design of the system
- Examples of benchmarks integrated into your framework

Your video presentation should give the judges an overview of:

- What your autoresearch system does
- What is novel or interesting about your approach
- What problems you tested it on
- What you would do next if you had more time

Judging Criteria

Submissions will be assessed across four areas:

1. Novelty and inventiveness

How original is the approach to automated research?

We are interested in creative approaches to how the system generates ideas, performs experiments, learns from results, coordinates agents, searches over possible solutions, or otherwise automates the research process.

2. Performance

How effectively does the system improve solutions on the benchmark task(s) it is applied to?

Demonstrated improvements from actual runs are strong evidence of performance. However, given the limited duration of the hackathon, judges will also consider the design of mechanisms intended to improve research performance, for example, how the system learns from previous experiments, selects promising directions, or avoids repeating unproductive searches.

3. Interpretability and ease of use

How easy is the framework to understand, run, and apply to new problems?

A strong submission should make it straightforward to understand the research process, inspect what the system discovered, reproduce experiments, and adapt the framework to another problem within its intended domain.

4. Research efficiency

How efficiently does the system use compute, experimental runs, and LLM tokens to make progress?

Compute and LLM Access

Each team will be provided with compute resources and LLM/API tokens for the duration of the hackathon.

Access instructions: [INSERT COMPUTE / TOKEN INSTRUCTIONS HERE]

Details on available models, usage limits, compute resources, and how to obtain credentials are provided in the instructions.

Where to Start

A simple autoresearch loop might look like:

Generate idea → implement candidate → run experiment → analyse result
→ retain useful discoveries → generate the next idea

But you are not restricted to this structure.

You are encouraged to experiment with the research process itself.

The goal is not simply to discover a good algorithm. The goal is to build a system that can discover good algorithms.

Contact us

Please reach out if you need help during the event.

Discord: <https://discord.gg/ZN36RsFPs>

Email: admin@algorithmdiscovery.org